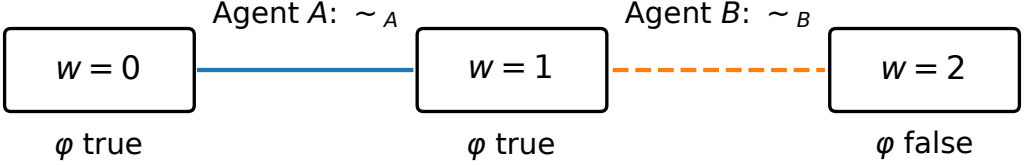

At $w = 0$: $K_A\varphi \ \wedge \ K_B\varphi$

But $0 \sim_A 1 \sim_B 2$, so $C_{\{A,B\}}\varphi$ is false.